

**“Heat Wave Response, Recovery, Future
Challenges,
and the Role of Technology: A Ground-
Level
Assessment and Preparedness Study”**



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ABSTRACT

Heat waves have become one of the most significant climate-related disasters affecting India due to rising global temperatures, rapid urbanization, deforestation, and changing land-use patterns. These extreme heat events pose serious threats to human health, agriculture, water resources, livelihoods, urban infrastructure, electricity demand, and public health systems. In recent years, states such as Telangana, Andhra Pradesh, Odisha, Rajasthan, Maharashtra, and Delhi have experienced frequent and severe heat waves, emphasizing the need for effective disaster preparedness and management.

This project, titled "**Heat Wave Response, Recovery, Future Challenges, and the Role of Technology: A Ground-Level Assessment and Preparedness Study,**" examines the causes and impacts of heat waves, evaluates existing response mechanisms, and analyzes recovery measures adopted after heat-related disasters. It also explores future challenges arising from climate change and highlights the importance of technological interventions in improving disaster preparedness and response.

The study is based on both primary and secondary data. Primary data includes field surveys, questionnaires, interviews, community interactions, and temperature observations, while secondary data is collected from the India Meteorological Department (IMD), National Disaster Management Authority (NDMA), government reports, research publications, and other reliable sources. The project also examines the role of modern technologies such as Artificial Intelligence (AI), Geographic Information Systems (GIS), Remote Sensing, Internet of Things (IoT), drones, satellite imagery, and mobile applications in heat wave prediction, monitoring, and mitigation.

The findings of this study are expected to enhance understanding of heat wave management, identify vulnerable populations, evaluate the effectiveness of existing preparedness measures, and recommend practical technological and policy improvements. The project contributes to strengthening climate resilience, promoting sustainable urban planning, improving public health protection, and supporting smart disaster management systems for reducing the impacts of future heat wave events.

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CHAPTER 1

Introduction to Heat Waves

A heat wave is a prolonged period of unusually high temperatures that is significantly above the normal climate of a region. Heat waves are among the most dangerous natural hazards because they affect large populations over several days and have serious impacts on health, agriculture, water resources, infrastructure, and the economy. In India, the India Meteorological Department (IMD) issues heat wave warnings based on temperature thresholds to help authorities and the public prepare for extreme weather conditions.

In recent years, the frequency and intensity of heat waves have increased due to climate change, rapid urbanization, deforestation, and changes in land-use patterns. The replacement of green spaces with buildings, roads, and concrete surfaces has intensified the Urban Heat Island (UHI) effect, causing cities to experience higher temperatures than surrounding rural areas.

Heat waves have become a major concern in India, particularly in states such as Telangana, Andhra Pradesh, Odisha, Rajasthan, Maharashtra, and Delhi. These extreme heat events result in dehydration, heat stroke, reduced agricultural productivity, water shortages, increased electricity demand, and pressure on public health systems. Vulnerable groups such as children, elderly people, outdoor workers, farmers, construction workers, and people with existing health conditions are at greater risk during heat waves.

Effective heat wave management requires timely forecasting, public awareness, emergency response systems, and long-term climate adaptation measures. Government agencies such as the India Meteorological Department (IMD) and the National Disaster Management Authority (NDMA) have developed Heat Action Plans to reduce the impact of extreme heat events. Modern technologies including Artificial Intelligence (AI), Geographic Information Systems (GIS), Remote Sensing, Internet of Things (IoT), drones, and mobile applications are also being used to improve heat wave prediction, monitoring, and preparedness.

This project studies heat wave response mechanisms, recovery measures, future challenges, and the role of technology in disaster management. It aims to understand the impacts of heat waves, identify vulnerable communities, evaluate existing response systems, and recommend practical measures to improve preparedness and reduce future risks.

1.2 Objectives of the Study

The main objective of this project is to understand the causes, impacts, response mechanisms, recovery measures, future challenges, and technological solutions related to heat waves in India. The study also aims to evaluate the preparedness of communities and disaster management agencies in reducing the adverse effects of extreme heat events.

Primary Objectives

- To understand the causes and impacts of heat waves.
- To analyze the response mechanisms adopted during heat waves.
- To study recovery measures implemented after heat-related disasters.
- To identify future challenges associated with rising temperatures and climate change.

Secondary Objectives

- To identify vulnerable populations affected by heat waves.
- To study the Urban Heat Island (UHI) effect.
- To evaluate the level of public awareness regarding heat wave safety.
- To suggest technological and policy improvements for effective heat wave management.

1.3 Scope of the Project

This project focuses on understanding heat wave disaster management through the study of response, recovery, preparedness, and technological interventions. The study can be conducted in urban areas, rural villages, municipal towns, schools, hospitals, industrial areas, and other heat-prone regions. It includes the collection of primary and secondary data through surveys, interviews, government reports, weather records, and research publications.

The project also examines the role of modern technologies such as Artificial Intelligence (AI), Geographic Information Systems (GIS), Remote Sensing, Internet of Things (IoT), drones, satellite imagery, and mobile applications in improving heat wave prediction, monitoring, and disaster preparedness. The findings of the study are intended to support better planning, increase public awareness, and strengthen climate resilience and disaster management practices.

1.4 Research Questions

The study is based on the following research questions:

1. What are the major impacts of heat waves on communities?
2. How effective are the existing heat wave response systems?
3. What challenges are expected in the future due to climate change?
4. How can technology improve heat wave preparedness and response?
5. Which communities are most vulnerable to heat wave impacts?

CHAPTER 2

Literature Review

2.1 Introduction

A literature review helps in understanding the existing knowledge, policies, and research related to heat wave management. It provides information on the causes, impacts, preparedness measures, and technological solutions adopted by government agencies and researchers. For this study, reports and guidelines published by national and international organizations were reviewed to understand current practices and recommendations for heat wave disaster management.

2.2 India Meteorological Department (IMD) Heat Wave Guidelines

The India Meteorological Department (IMD) is responsible for monitoring weather conditions and issuing heat wave forecasts and alerts across India. It defines heat wave conditions based on temperature thresholds and regional climatic variations. IMD provides early warnings that help government departments, disaster management authorities, healthcare services, and the public take preventive measures to reduce heat-related risks.

2.3 National Disaster Management Authority (NDMA) Heat Action Plan

The National Disaster Management Authority (NDMA) has developed Heat Action Plan guidelines to minimize the impacts of extreme heat. These guidelines recommend early warning systems, public awareness campaigns, availability of drinking water, cooling shelters, medical preparedness, and coordination among government departments. The objective is to reduce heat-related illnesses and deaths through effective planning and response.

2.4 Telangana State Heat Action Plan

The Telangana State Heat Action Plan focuses on protecting vulnerable populations during extreme heat events. The plan emphasizes public awareness, continuous weather monitoring, emergency healthcare services, safe drinking water facilities, and coordination between district authorities and disaster management agencies. These measures improve preparedness and reduce the impact of heat waves across the state.

2.5 IPCC and WHO Reports

The Intergovernmental Panel on Climate Change (IPCC) states that climate change has increased the frequency and intensity of heat waves worldwide. The World Health Organization (WHO) highlights that prolonged exposure to extreme heat can cause dehydration, heat exhaustion, heat stroke, cardiovascular problems, and increased mortality, particularly among elderly people, children, and individuals with pre-existing health conditions.

2.6 Urban Heat Island Studies

Urban Heat Island (UHI) studies show that cities experience higher temperatures than surrounding rural areas because of concrete buildings, roads, reduced vegetation, and increased human activities. Expanding green spaces, planting trees, and adopting cool roofs and reflective materials are effective strategies to reduce urban temperatures.

2.7 Summary

The reviewed literature indicates that heat waves are becoming more frequent due to climate change and urbanization. Effective disaster management requires accurate forecasting, public awareness, inter-departmental coordination, and the use of modern technologies such as AI, GIS, Remote Sensing, IoT, and mobile applications. These studies provide a strong foundation for understanding heat wave preparedness, response, and long-term mitigation strategies.

CHAPTER 3

Study Area Profile

3.1 Study Area

The study was conducted with reference to **Hyderabad, Telangana**, one of the rapidly growing metropolitan cities in southern India. Hyderabad has experienced increasing temperatures in recent years due to rapid urbanization, population growth, reduced green cover, and changing climatic conditions. The city frequently records high summer temperatures, making it suitable for studying heat wave impacts and disaster management practices.

3.2 Geographic Location

Hyderabad is located on the Deccan Plateau in the state of Telangana. The city lies at an average elevation of about **542 metres above sea level** and has a tropical wet and dry climate. Summers generally extend from March to June, during which temperatures often exceed **40°C**.

3.3 Climate and Temperature

The climate of Hyderabad is characterized by hot summers, a monsoon season from June to September, and mild winters. During severe heat wave conditions, maximum temperatures may reach **42–45°C**. High temperatures combined with humidity and urban development increase the risk of heat-related illnesses.

3.4 Heat Wave Vulnerability

Several factors make Hyderabad vulnerable to heat waves:

- Rapid urbanization and expansion of built-up areas.
- Reduction in vegetation and green spaces.
- High population density.
- Heavy traffic and industrial activities.
- Increasing demand for water and electricity during summer.
- Large number of outdoor workers exposed to extreme heat.

3.5 Importance of the Study Area

Hyderabad was selected as the study area because it represents a rapidly developing urban environment where climate change and urban heat island effects have increased heat-related risks. Studying this area helps in understanding existing preparedness measures, identifying vulnerable groups, and recommending suitable technological and policy interventions for effective heat wave management.

CHAPTER 4

Research Methodology

4.1 Research Design

This study adopts a descriptive research approach to understand the causes, impacts, response mechanisms, recovery measures, future challenges, and technological solutions related to heat waves. The research combines both primary and secondary data to provide a comprehensive assessment of heat wave disaster management.

4.2 Data Collection

The study uses the following methods for data collection:

Primary Data

- Field surveys
- Questionnaires
- Interviews
- Temperature observations
- Community interaction
- Focus group discussions

Secondary Data

- India Meteorological Department (IMD) temperature records
- National Disaster Management Authority (NDMA) guidelines
- Government reports
- Research papers and journals
- Municipal records
- Health department reports

4.3 Sampling Areas

The study focuses on areas and communities that are more vulnerable to extreme heat conditions. The selected groups include:

- Traffic police
- Street vendors
- Farmers
- Construction workers
- Elderly people
- Children
- Slum residents
- Sanitation workers

4.4 Tools Used

The following tools and technologies are used during the study:

- Thermometer
- GPS Mapping
- Mobile Survey Applications
- Geographic Information System (GIS)

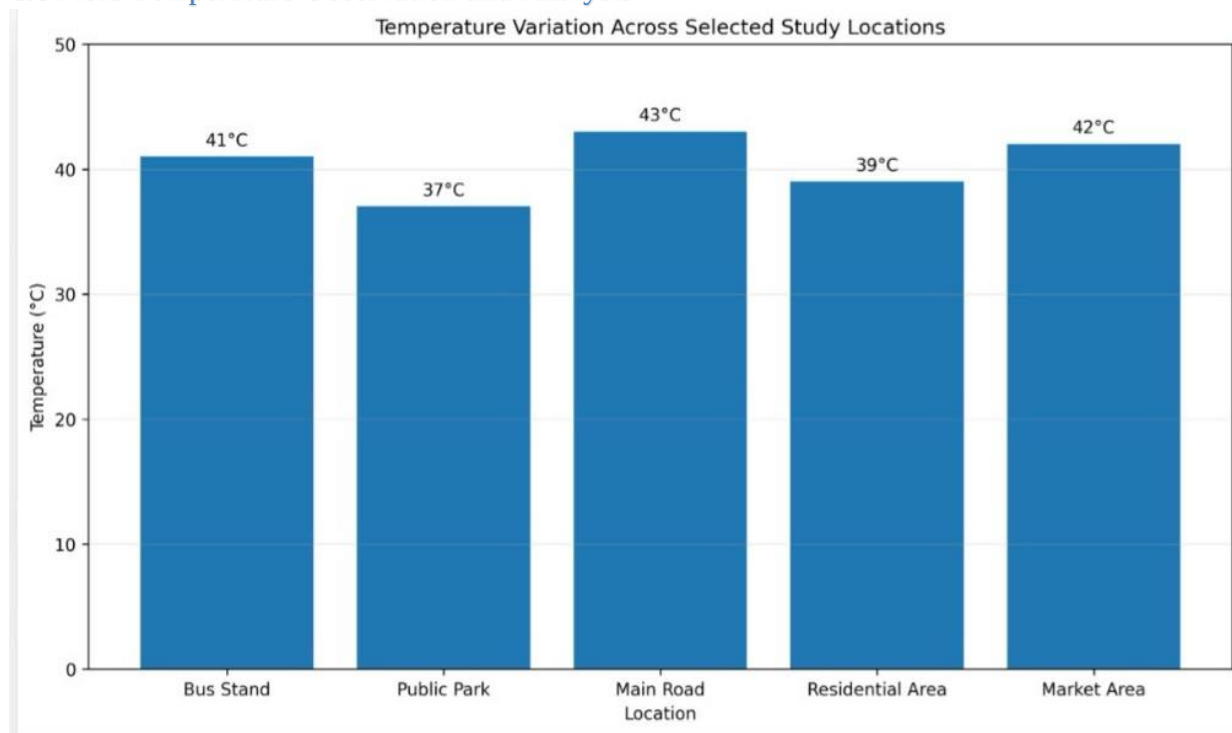
- Satellite Imagery
- Google Earth
- IMD Weather Data

4.5 Research Process

The project was carried out through the following steps:

1. Literature review on heat waves and disaster management.
2. Collection of primary and secondary data.
3. Identification of vulnerable groups and study areas.
4. Analysis of heat wave response and recovery measures.
5. Assessment of technological applications for heat wave preparedness.
6. Preparation of findings, recommendations, and final report.

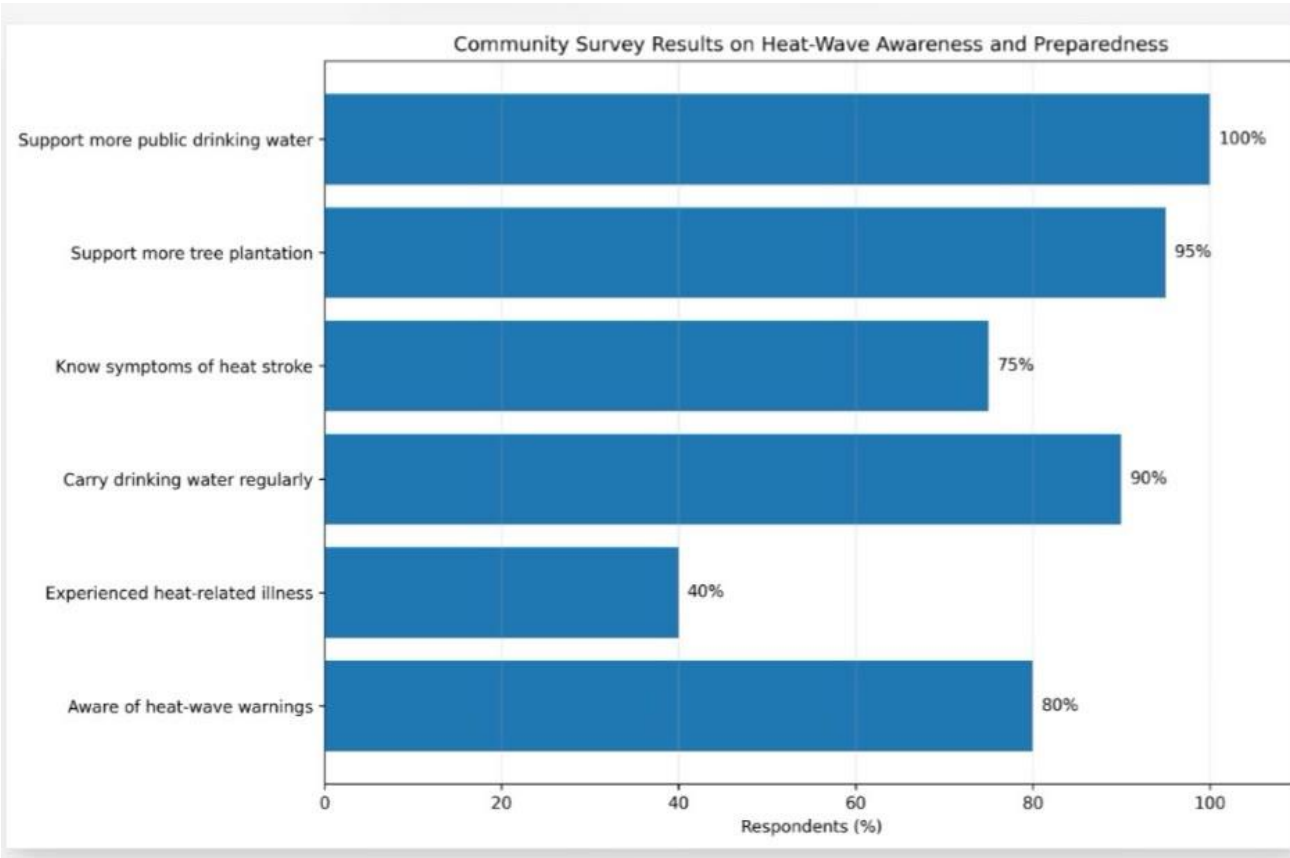
4.6 Field Temperature Observation and Analysis



4.7 Community Survey Analysis

- A community survey involving 20 participants was conducted to understand awareness, preparedness, and experiences related to heat waves. The survey focused on awareness of heat-wave warnings, heat-related illness, personal safety practices, knowledge of heat-stroke symptoms, and support for local heat mitigation measures.
- The results showed that **16 participants (80%)** were aware of heat-wave warnings, while **8 participants (40%)** reported experiencing heat-related illness. **18 participants (90%)** regularly carried drinking water, and **15 participants (75%)** were aware of the symptoms of heat stroke.
- Support for preventive measures was particularly high. **19 participants (95%)** supported increased tree plantation, while **all 20 participants (100%)** supported the installation of additional public drinking-water facilities.
- Overall, the survey indicates a relatively good level of heat-wave awareness among participants. However, the occurrence of heat-related illness and gaps in knowledge of heat-stroke symptoms show the need for continued public awareness campaigns, improved shade facilities, and greater access to drinking water.

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4.8 Heat Vulnerability Assessment

The observed temperature differences across the selected locations were used to assess relative heat exposure. Areas with higher temperatures, heavy traffic, dense built-up surfaces, limited vegetation, and fewer shaded spaces were considered comparatively more vulnerable to heat stress.

The Main Road recorded the highest temperature at 43°C, followed by the Market Area at 42°C and the Bus Stand at 41°C. These locations represent areas with greater human activity, traffic, and exposure to built surfaces. The Residential Area recorded 39°C, indicating moderate heat exposure, while the Public Park recorded the lowest temperature at 37°C, demonstrating the cooling effect associated with vegetation and shade.

Based on these observations, the locations can be classified as:

Study Location	Temperature	Relative Heat Exposure
Main Road	43°C	Very High
Market Area	42°C	High
Bus Stand	41°C	High

Residential Area	39°C	Moderate
Public Park	37°C	Low

This comparison suggests that urban greenery and shaded spaces can play an important role in reducing local heat exposure, while highly built-up and traffic-intensive areas require greater attention during heat-wave preparedness and response.

CHAPTER 5

Heat Wave Response Analysis

5.1 Introduction

Heat wave response refers to the immediate actions taken by government agencies, disaster management authorities, healthcare services, and local communities to reduce the impact of extreme heat. A timely and coordinated response helps protect human lives, minimize health risks, and ensure the availability of essential services during heat wave conditions.

5.2 Public Warning Systems

The India Meteorological Department (IMD) issues heat wave forecasts and alerts through television, radio, newspapers, websites, mobile applications, and SMS notifications. These early warning systems help government departments and the public prepare for extreme weather conditions.

5.3 Cooling Centres and Drinking Water Facilities

During severe heat waves, local authorities establish cooling centres in public buildings where people can rest and avoid direct exposure to high temperatures. Public drinking water facilities are arranged at bus stations, markets, hospitals, schools, and other crowded places to reduce dehydration and heat-related illnesses.

5.4 Medical and Emergency Response

Hospitals and healthcare centres remain prepared to treat patients suffering from dehydration, heat exhaustion, and heat stroke. Ambulance services are kept on alert to provide immediate medical assistance. Health departments also advise people to avoid outdoor activities during peak afternoon hours.

5.5 Community Awareness and Role of District Administration

Public awareness campaigns are conducted through newspapers, television, social media, and community meetings to educate people about heat wave safety measures. District administrations coordinate with health departments, municipal authorities, disaster management agencies, and emergency services to ensure effective implementation of Heat Action Plans.

5.6 Response Indicators

The effectiveness of heat wave response can be assessed using the following indicators:

- Availability of drinking water facilities.
- Availability of shaded public areas.
- Timely dissemination of heat alerts.
- Accessibility of emergency healthcare services.
- Community awareness regarding heat safety.
- Coordination among government departments.

5.7 Summary

An effective heat wave response system depends on timely forecasting, public awareness, emergency healthcare, adequate water supply, and coordinated action by disaster management authorities. Strengthening these measures can significantly reduce heat-related illnesses, economic losses, and fatalities during extreme heat events.

5.8 Case Study: Heat-Wave Preparedness in Hyderabad

Hyderabad is a rapidly growing metropolitan city where high summer temperatures, dense construction, heavy traffic, reduced vegetation, and increasing population contribute to heat-wave vulnerability. The risk is particularly significant for people who spend long periods outdoors, including street vendors, construction workers, traffic personnel, sanitation workers, and daily-wage workers.

Heat-wave preparedness in the city depends on timely weather warnings, access to drinking water, shaded public spaces, public awareness, and healthcare preparedness. Early warnings issued by the India Meteorological Department help residents and authorities prepare for periods of extreme heat. Local-level measures such as providing drinking-water facilities, reducing outdoor exposure during peak afternoon hours, and spreading information about dehydration and heat-stroke symptoms can further reduce health risks.

The field observations in this study demonstrate differences in heat exposure across urban environments. The Main Road recorded **43°C**, compared with **37°C at the Public Park**, a difference of 6°C. The Market Area and Bus Stand also recorded relatively high temperatures of **42°C and 41°C**, respectively. These observations suggest that areas with heavy traffic, concrete surfaces, crowding, and limited shade may experience greater heat exposure, while vegetation and tree cover can provide localized cooling.

Technology can further strengthen preparedness through GIS-based hotspot identification, satellite monitoring, automated temperature sensors, mobile heat alerts, and AI-assisted forecasting. Combining these technologies with public awareness, urban greenery, drinking-water facilities, and targeted protection for vulnerable populations can improve Hyderabad's resilience to future heat-wave events.

CHAPTER 6

Recovery Measures

6.1 Introduction

Recovery refers to the process of restoring normal life after the impacts of a heat wave. It includes improving public health, restoring livelihoods, ensuring water and electricity supply, supporting agriculture, and reducing the long-term effects of extreme heat on communities.

6.2 Health Recovery

People affected by heat-related illnesses require timely medical treatment, proper hydration, and

continuous monitoring. Hospitals and healthcare centres play a vital role in treating patients suffering from dehydration, heat exhaustion, and heat stroke. Public awareness regarding heat safety also helps reduce future health risks.

6.3 Economic and Livelihood Recovery

Heat waves affect daily wage workers, farmers, street vendors, construction workers, and other outdoor workers by reducing working hours and productivity. Recovery measures include providing financial support, restoring employment opportunities, and implementing welfare schemes for affected communities.

6.4 Agricultural and Water Resource Recovery

High temperatures can reduce crop yields, affect livestock, and increase irrigation requirements. Recovery measures include efficient water management, improved irrigation practices, crop protection, and conservation of water resources to support agricultural activities after heat wave events.

6.5 Recovery Indicators

The effectiveness of recovery can be evaluated using the following indicators:

- Number of affected persons.
- Hospital admissions due to heat-related illnesses.

- Number of workdays lost.
- Crop losses.
- Water scarcity levels.
- Electricity supply restoration.

6.6 Summary

Effective recovery measures help communities return to normal life after heat wave events. Strengthening healthcare services, protecting livelihoods, conserving water resources, and improving disaster management planning can reduce the long-term impacts of future heat waves.

CHAPTER 7

Future Challenges

7.1 Introduction

Climate change is increasing the frequency, duration, and intensity of heat waves across India. Rising temperatures, rapid urbanization, and environmental changes are creating new challenges for disaster management agencies and communities. Addressing these challenges requires long-term planning, sustainable development, and the use of modern technology.

7.2 Major Future Challenges

The major challenges expected in the coming years include:

- **Climate Change:** Rising global temperatures are expected to increase the occurrence and severity of heat waves.
- **Urban Heat Island Effect:** Rapid urbanization, loss of green spaces, and extensive concrete structures increase temperatures in cities.
- **Water Shortages:** Higher temperatures increase water demand while reducing the availability of freshwater resources.
- **Population Growth:** Growing populations place additional pressure on water, electricity, healthcare, and public infrastructure.
- **Increasing Energy Demand:** The use of air conditioners, coolers, and other cooling devices increases electricity consumption during summer.
- **Health Emergencies:** Heat-related illnesses such as dehydration, heat exhaustion, and heat stroke are expected to become more common, particularly among vulnerable groups.
- **Migration Due to Heat Stress:** Extreme heat may force people to migrate from severely affected areas in search of safer living conditions and employment.
- **Reduced Labour Productivity:** Outdoor workers, including farmers, construction workers, and street vendors, may experience reduced working hours and income due to unsafe temperatures.
- **Heat Stress in Animals:** High temperatures also affect livestock and wildlife, reducing productivity and increasing health risks.
- **Infrastructure Challenges:** Roads, power systems, water supply networks, and public infrastructure may experience increased stress and damage during prolonged heat waves.

7.3 Need for Future Preparedness

To reduce the impacts of future heat waves, governments and communities should strengthen Heat Action Plans, increase urban greenery, improve water conservation, promote climate-resilient infrastructure, and encourage the use of advanced technologies for early warning and disaster preparedness.

7.4 Summary

Future heat wave risks can be minimized through effective planning, sustainable urban development, public awareness, and technological innovation. Strengthening disaster management systems today will help communities become more resilient to extreme heat events in the future.

CHAPTER 8

Role of Technology

8.1 Introduction

Technology plays a vital role in reducing the impact of heat waves by improving forecasting, monitoring, preparedness, and emergency response. Modern technologies enable government agencies to issue timely warnings, identify high-risk areas, monitor temperature changes, and support effective disaster management.

8.2 Early Warning Systems

The India Meteorological Department (IMD) provides weather forecasts and heat wave alerts using advanced meteorological models. These alerts are communicated through SMS, mobile applications, television, radio, and social media, helping people take preventive measures before extreme heat conditions occur.

8.3 Geographic Information System (GIS) and Remote Sensing

GIS and Remote Sensing technologies are used to identify heat-prone areas, prepare thermal hotspot maps, monitor land surface temperatures, and support urban planning. Satellite imagery helps disaster management authorities understand temperature variations and identify vulnerable locations.

8.4 Artificial Intelligence (AI)

Artificial Intelligence is used to analyze weather data and predict future heat wave events. AI-based systems support risk assessment, identify vulnerable populations, and assist authorities in planning timely response measures.

8.5 Internet of Things (IoT)

IoT devices such as smart temperature sensors and automated weather stations continuously monitor environmental conditions. These systems provide real-time information that helps authorities respond quickly during extreme heat events.

8.6 Drones and Mobile Applications

Drones can be used to monitor vulnerable areas, assess disaster situations, and support emergency response activities. Mobile applications provide weather updates, heat alerts, safety guidelines, and disaster reporting facilities, enabling people to stay informed during heat waves.

8.7 Smart Infrastructure

Smart infrastructure helps reduce heat stress in urban areas. Measures such as cool roofs, reflective pavements, green buildings, urban green spaces, and smart irrigation systems improve thermal comfort, conserve energy, and increase climate resilience.

8.8 Summary

The integration of technologies such as AI, GIS, Remote Sensing, IoT, drones, mobile applications, and smart infrastructure has significantly improved heat wave prediction, preparedness, and response. Continued investment in these technologies will strengthen disaster management systems and reduce the impact of future heat wave events.

CHAPTER 9

Findings and Recommendations

9.1 Findings

Based on the study of heat wave response, recovery measures, future challenges, and the role of technology, the following key findings were observed:

- Heat waves are becoming more frequent and intense due to climate change and rapid urbanization.
- Vulnerable groups such as elderly people, children, outdoor workers, farmers, street vendors, and construction workers are at the highest risk of heat-related illnesses.
- Early warning systems issued by the India Meteorological Department (IMD) help reduce the impact of heat waves when communicated effectively.
- Public awareness regarding heat wave safety has improved, but many communities still lack sufficient knowledge about preventive measures.
- Water scarcity and increased electricity demand are major challenges during extreme heat events.
- Modern technologies such as Artificial Intelligence (AI), Geographic Information Systems (GIS), Remote Sensing, Internet of Things (IoT), drones, and mobile applications improve heat wave forecasting, monitoring, and disaster preparedness.
- Effective coordination among government departments, healthcare services, and disaster management authorities is essential for reducing heat-related risks.

9.2 Recommendations

The following recommendations are suggested to improve heat wave preparedness and disaster management:

- Increase urban greenery through tree plantation and the development of parks.
- Strengthen district-level Heat Action Plans and ensure effective implementation.
- Improve early warning systems and ensure timely dissemination of heat alerts.
- Install public drinking water facilities and shaded rest areas in busy public locations.
- Promote cool roof technology, reflective pavements, and green buildings to reduce urban temperatures.
- Use Artificial Intelligence (AI) and Geographic Information Systems (GIS) for temperature forecasting and heat risk assessment.
- Conduct regular public awareness campaigns on heat wave safety and first aid.
- Strengthen coordination among disaster management authorities, health departments, municipal bodies, and emergency services.
- Encourage water conservation and sustainable urban planning to reduce future heat-related risks.

9.3 Summary

The findings indicate that effective planning, community awareness, technological innovation, and coordinated disaster management can significantly reduce the impacts of heat waves.

Implementing the above recommendations will improve preparedness, protect vulnerable populations, and strengthen climate resilience in the future.

CHAPTER 10

Conclusion

Heat waves have emerged as one of the most significant climate-related disasters affecting India. Increasing temperatures caused by climate change, rapid urbanization, and environmental degradation have increased the frequency and intensity of extreme heat events. These events have serious impacts on human health, agriculture, water resources, livelihoods, urban infrastructure, and public health systems.

This study examined the causes and impacts of heat waves, existing response mechanisms, recovery measures, future challenges, and the role of modern technology in disaster management. The findings indicate that effective preparedness, timely early warning systems, public awareness, coordinated emergency response, and sustainable recovery measures can significantly reduce heat-related risks.

The study also highlights the importance of technologies such as Artificial Intelligence (AI), Geographic Information Systems (GIS), Remote Sensing, Internet of Things (IoT), drones, and smart infrastructure in improving heat wave prediction, monitoring, and management. These technologies support better decision-making and enhance the overall efficiency of disaster management systems.

In conclusion, strengthening Heat Action Plans, increasing community awareness, promoting sustainable urban development, and adopting advanced technologies will improve climate resilience and help protect vulnerable populations from future heat wave disasters. Continuous collaboration among government agencies, researchers, healthcare institutions, and local communities is essential for building a safer and more resilient society.

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APPENDIX

Appendix A – Sample Survey Questionnaire

The following questionnaire was prepared to understand public awareness, preparedness, and experiences related to heat waves.

1. What is your age?
2. What is your occupation?
3. Are you aware of heat wave warnings issued by the India Meteorological Department (IMD)?
4. Have you ever experienced health problems due to extreme heat?
5. What precautions do you take during a heat wave?
6. Is safe drinking water easily available in your area during summer?
7. Are there enough shaded public places near your residence or workplace?
8. Do you know the symptoms of heat stroke?
9. Have you received any heat wave awareness information from the government or local authorities?
10. What measures do you suggest to reduce the impact of heat waves in your area?

Appendix B – Sample Temperature Observation Table

Location	Date	Time	Temperature (°C)	Observation
Bus Stand	20-07-2026	1:00 PM	41	High pedestrian movement, very hot
Public Park	20-07-2026	2:00 PM	37	Cooler due to trees
Main Road	20-07-2026	3:00 PM	43	Heavy traffic and high heat
Residential Area	20-07-2026	4:00 PM	39	Moderate temperature
Market Area	20-07-2026	12:00 PM	42	Crowded and limited shade

Appendix C – Sample Survey Summary

Number of Participants: 20

- Aware of heat wave warnings: 16
- Experienced heat-related illness: 8
- Carry drinking water regularly: 18
- Know symptoms of heat stroke: 15
- Support more tree plantation: 19
- Support additional public drinking water facilities: 20

Appendix C.1 – Survey Data Summary

Survey Indicator	Yes	No	Yes (%)
Aware of heat-wave warnings	16	4	80%
Experienced heat-related illness	8	12	40%
Carry drinking water regularly	18	2	90%
Know symptoms of heat stroke	15	5	75%
Support more tree plantation	19	1	95%
Support additional public drinking-water facilities	20	0	100%

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Appendix D – Suggested Figures

Green/Shaded Area.



Exposed Built-Up Area



Drinking Water Facility



Traffic/High-Exposure Area

